# Who Returns the Ballot?

### Cross-state evidence on age, party, and low-salience electorates — individual-record data from Washington, New York, and Idaho

**DRAFT — pending human/editorial sign-off.** That gate is a person reading the paper end to end, recorded in `who-returns-ballot-submission-notes.md` §Sign-off; `scripts/verify_who_returns_ballot.py` asserts the figures and is not the sign-off. First assembly 2026-07-22; harmonized and scoped to three states 2026-08-06. Prose frame, case design, and harmonization protocol are complete. **Findings 1 and 3 are computed from a single code path** — `scripts/diag_cross_state_age_harmonized.py` — over the three voter files, with shared age bins, age measured at each election date, and one ACS vintage for every state's CVAP benchmark. They are not the three single-state papers' as-reported figures.

**This is a three-state paper by decision, not by omission** (2026-08-06). An earlier draft described a four-state design with Texas deferred. Texas is now stated as *future work* in Boundary of inference rather than as a gap in this one: the claim here is about the gradient within each state's own salience ladder, and three states already span the axes the design turns on — a high-access no-party-registration state, a blue party-registration state, and a red closed-primary state. What Texas would add is named precisely, so a reader can judge the cost of its absence rather than discover it. **The one remaining blocker is the standing human-verification gate.**

Author: Stephen Kirby (Tikor). Companion to the SSRN-posted lead paper *Who Decides Washington?* (SSRN Abstract ID 7149263).

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## The question

A single-state paper answers *what does this state's low-salience electorate look like?* A multi-state paper answers the general-behavior question: *how far does the age gradient across lower-salience electorates travel, and how does it change with party-registration rules and election administration?* This paper takes the validated, individual-record method of the Washington lead paper and runs it identically across three states chosen to differ on the axes that should matter — a high-access no-party-registration state (WA), a blue party-registration state (NY), and a red party-registration state (ID) — to separate the part of the pattern that recurs across all three cases from the part that is a feature of a particular state's institutions.

The two questions this state set can answer, and the one it cannot:

1. Is the age gradient across lower-salience electorates present across blue, red, and mixed institutional settings?
2. In party-registration states, does participation in the lowest-salience measured contest skew toward one party? ("Off-cycle" is not the right word for that contest everywhere — Idaho's is a closed May primary; see the salience note below.)
3. *Not answerable here.* Where there is no party of record, does the age skew still have partisan consequences? Washington is the set's only no-party-registration state and its top-two primary records no party ballot, so nothing in this data can resolve it. A state with no party of record but a **public party-primary history** — Texas is the obvious one — would supply the proxy that makes the question tractable. That is named as future work in Boundary of inference, not carried as a placeholder in the design.

## The cases

| State | Party of record | What it contributes | Data on hand |
|---|---|---|---|
| **Washington** | No | High-access baseline: universal vote-by-mail, same-day registration, automatic registration. The clean "access is easy" case; age only. | `data/wa_vrdb.duckdb` (5.51M; year of birth) |
| **New York** | **Yes** | Cleanest blue party-registration comparison. | `data/ny_vrdb.duckdb` (13.54M; party + year of birth) |

| State | Party of record | What it contributes | Data on hand |
|---|---|---|---|
| **Idaho** | **Yes** | Red party-registration comparison; a *closed* primary makes party determine which election a voter may even enter. | `data/id_vrdb.duckdb` (1.03M; party + age + primary ballot) |

The three cases were chosen to vary the two institutional axes the argument turns on — whether the state records party at all, and whether its lowest-salience measured contest is open to everyone — while holding the computation protocol fixed. What is held fixed is the harmonization of the next section — the estimand, the age bands, age measured at each election date, one ACS vintage — not measurement itself: contest type, party variable, survivorship regime and the meaning of the lowest rung all differ by state, and Boundary of inference prices each in. They do not vary a third axis: **no state here lacks a party of record *and* publishes a party-primary history**, which is the configuration that would let the partisan question be asked where party is unrecorded. Texas is that case and is not loaded; see Boundary of inference.

## Harmonization protocol — the load-bearing methods problem

"Off-year" does not mean the same thing in every state, and pretending it does would manufacture a false equivalence. This paper therefore compares **election *classes* defined by salience**, not by calendar:

 **"Salience" here is a RESEARCHER-IMPOSED ORDINAL, not a measured variable, and the ranking below is the paper's own construction.** Nothing in the data scores an election's salience; the ladder orders classes by an assumed prominence and the analysis then reports composition within each. Idaho's class is the sharpest illustration of why the label must not be read as a single dimension: its lowest rung is simultaneously lower-turnout, partisan, eligibility- restricted, a nomination rather than a final contest, and differently timed. Calling its distance from the presidential general a *salience* gradient would give a latent construct causal weight the design cannot support — which is why the heading of Finding 1 says "age gradient across lower-salience electorates" and not "off-cycle age skew", corrected 2026-08-26, and why what the paper concludes is the WITHIN-STATE statement: every selected lower-salience electorate is older than its own state's presidential electorate.

- **Highest-salience** — the presidential November general. Directly comparable across all three states.
- **Midterm** — the even-year non-presidential November general. Directly comparable.
- **Lowest-salience measured contest** — *state-specific by construction, and reported as such*:
- **WA:** the odd-year November local general (RCW 29A.04.321).
- **NY:** the odd-year November local general.
- **ID:** there is no odd-year November general; the lowest-salience measured contest is the **closed May Republican primary** — the dominant-party nominating electorate.

*This class was called Idaho's "decisive" contest until 2026-08-17, and that label is withdrawn. Whether — and in how many seats — the Republican primary settles the outcome is a question about seat decision, not about salience or composition, and this paper measures neither. The Idaho companion separates nomination, filing, primary and general as distinct stages and declines to call the primary decisive, on its own state's November results — see `who-decides-idaho.md` §IV–§V, which owns those figures and gates them. Nothing in Findings 1 or 3 depends on the label: the contest enters this paper as the lowest-salience electorate each state's file records, and it would enter identically if it decided nothing.*

Each state's lowest-salience row is labeled with its class so the comparison is never read as a shared calendar. Three further comparability controls travel with every figure:

- **Age precision differs** (WA year-of-birth; NY/ID date-of-birth or integer age) — all cuts use whole-year age bins to a common definition.
- **Vote-history depth differs** — metrics use each state's available reconstructed history and state the window.
- **List-maintenance differs** — Idaho's roll contracted ~13% over the window, which makes ID turnout *rates* unusable; ID is therefore reported as **composition shares only**, and all states foreground shares over rates (the WA paper's practice).

## Finding 1 — The age gradient across lower-salience electorates recurs across all three cases

As salience falls, the electorate ages, in each of the three states and regardless of party-registration regime. The presidential and midterm rows below are directly comparable; the lowest-salience row is each state's lowest-salience measured class (labeled).

| State | Presidential 65+ / 18–29 | Midterm 65+ / 18–29 | Lowest-salience 65+ / 18–29 (class) |
|---|---|---|---|
| **WA** | 28.5% / 14.2% | 31.0% / 10.4% | 36.8–40.3% / 7.4–8.0% (3 odd-year Nov generals) |
| **NY** | 28.2% / 14.1% | 30.6% / 9.8% | 28.4–41.6% / 6.0–11.5% (5 odd-year Nov generals) |
| **ID** | 29.0% / 15.2% | 34.4% / 8.6% | 41.4–48.6% / 5.0–5.6% (3 closed May GOP primaries)[1] |

[1] Idaho has no odd-year November general; its lowest-salience measured contest is the closed May Republican primary, whose electorate is older than any general (Idaho paper §IV). The midterm column already shows the gradient reaching 34.4% 65+ before the primary cut. All cells above are computed by `scripts/diag_cross_state_age_harmonized.py` from one code path over the three voter files — shared age bins, age measured at each election date — so they are directly comparable to each other rather than transcribed from three papers' per-state conventions. **The low-salience column uses every such contest in each state's loaded history**, not a selection: 3 / 5 / 3 contests. An earlier version used 3 / 2 / 1, and the omissions were doing work — New York's floor read 34.4% rather than 28.4%, and Idaho appeared as a single point rather than a range.

**The low-salience column is confounded with distance from the 2026 roll, and the confound is large.** These are current-roll reconstructions, so a voter who cast a ballot in an earlier contest appears only if they are still registered in 2026 — and departures skew old. Both multi-contest series run in exactly that direction:

| | **earliest → latest** |
|---|---|
| NY odd-year generals, 65+ | **28.4** (2017) → 32.9 (2019) → 36.3 (2021) → **41.6** (2023) → *34.4* (2025) |
| ID closed R primaries, 65+ | **41.4** (2022) → 46.7 (2024) → **48.6** (2026) |

**New York's 2025 cell is printed in italics because it breaks the pattern, and an earlier version of this table omitted it.** A recency ladder that drops its most recent observation, in a paragraph arguing recency drives the pattern, is the omission of the disconfirming case from the display that establishes the confound. 2025 is genuinely a different construct from the other four — it POSTDATES the November 2024 reference, so its reconstruction bias is near zero rather than growing with lag — but that is a reason to label it, not to leave it out. Read the first four as the lag series and 2025 as the control: the least-biased cell in the row reads younger than three of the four that precede it.

New York's odd-year 65+ share climbs 13 points across four contests in near-perfect order of recency, and Idaho's climbs 7 across three. Some of that is real — 2023 and 2025 were different contests — but no reading of these series should treat the spread as purely substantive, and **cross-state comparisons of the low-salience *level* are not safe**, because the states' cells sit at different lags (Idaho's 2024 primary is 2 years before the snapshot; New York's cells span 1 to 9). What the confound does not touch is the within-state *direction* — every low-salience contest is older than its own presidential one — which is the claim Finding 1 actually makes. The 2025 exception to New York's ladder is discussed below.

The three presidential rows are strikingly similar (28–29% 65+, 14–15% 18–29): the states' electorates *start* from nearly the same age structure at high salience and diverge only as salience falls, which is the signature of a turnout-composition (surge-and-decline) mechanism rather than a registration-age artifact. The Washington and New York papers' Kitagawa/Das Gupta-style decompositions attribute the 65+ rise mostly to differential turnout, not to the age structure of the roll; the Idaho paper reports its decomposition as **directionally consistent only** — its roll churn makes the numeric split unreliable there, and that paper deliberately prints no decomposition figures.

**How far the skew travels is *not* uniform, and the harmonized numbers are what show it.** The senior-to-youth ratio starts at almost exactly 2:1 in all three states (WA 2.0, NY 2.0, ID 1.9) and then diverges sharply: to **4.9–5.5:1** in Washington, **3.0–7.0:1** in New York, and **7.4–9.7:1** in Idaho's closed primaries. So the gradient runs the same way in all three cases in *direction* — every state's lowest-salience electorate is older than its presidential one — while its *magnitude* ranges from a 1.5× to a 5.1× multiplication of the ratio. (These ratios are computed from unrounded cohort shares, so they need not equal the quotient of the rounded percentages printed in the table above — the same convention the Idaho paper states for its R−D margins.)

Two consequences follow, and both are only visible once the metrics are computed identically:

- **Idaho's closed primary is the most age-unrepresentative low-salience electorate measured here**, not merely "grayest of all" as an earlier draft of this table put it. Its three primaries run 41.4–48.6% over 65 against 5.0–5.6% under 30, and its dissimilarity index (25.0–27.8) is above every general measured here. The claim is about the *class*: its

weakest cell, the 2022 primary at 41.4%, sits just below New York's 2023 general at 41.6%, so "exceeds every odd-year general" — which an earlier version asserted — is false on the full set and is withdrawn.

- **"Odd-year" is not one salience level.** New York's five odd-year generals span 28.4% to 41.6% over 65, a range of **13.2 points** — as wide as the gap between New York's presidential electorate and Idaho's *weakest* primary (13.2), though well short of the gap to its strongest (20.4). An earlier version called this range "wider than separates New York's presidential electorate from Idaho's closed primary"; on unrounded values it exceeds the narrowest of those separations by **0.05 points** and is smaller than every other reading of it. That is not a difference this measurement can resolve: the Idaho age-convention bracket alone is an order of magnitude larger (see Methods & reproducibility). The comparison is restated rather than withdrawn, because the point it makes does not need it. Part of the spread is also **window length**: New York's series covers nine years and Washington's five, so New York has more room to vary before any substantive heterogeneity. 2025 carried a high-salience New York City mayoral contest, which is one explanation for its being the youngest recent cell; note that **2021 was also a mayoral year and reads 36.3%** — and so was **2017, at 28.4%**, since New York City runs on a 2013/2017/2021/2025 cycle. So all three mayoral cells (28.4, 36.3, 34.4) sit below the two non-mayoral ones (32.9, 41.6), and "mayoral years run younger" covers both ends of the ladder that lag is otherwise asked to explain. With five contests neither reading is testable and the paper asserts neither; naming only one would be the error. Either way this is a caution against treating the calendar as a proxy for salience — the very substitution this paper's harmonization protocol was written to avoid.

## Finding 2 — The *partisan* structure of the skew is state-specific

Where we can see party (NY, ID), the age skew has sharply different partisan consequences — and this is the paper's general-vs-particular payoff:

**The comparison has to be made within a class, and doing so changes the reading.** An earlier version put New York's *odd-year* gap beside Idaho's *presidential* one — different rungs of the salience ladder this paper exists to build. On matched classes:

| | Republican 65+ − Democratic 65+ |
|---|---|
| NY, Nov 2016 presidential | **+2.7** pts (22.5 vs 19.8) |
| NY, Nov 2020 presidential | **+4.1** pts (26.8 vs 22.7) |
| NY, Nov 2024 presidential | **+3.7** pts (32.4 vs 28.7) |
| NY, Nov 2017 odd-year (NYC mayoral) | **+1.2** pts (30.0 vs 28.8) |
| NY, Nov 2019 odd-year | **+3.7** pts (35.8 vs 32.1) |
| NY, Nov 2021 odd-year (NYC mayoral) | **+1.0** pts (37.5 vs 36.6) |
| NY, Nov 2023 odd-year (its lowest-salience contest) | **+2.0** pts (43.5 vs 41.6) |
| NY, Nov 2025 odd-year (NYC mayoral) | **+10.7** pts (42.8 vs 32.1) |
| ID, Nov 2024 presidential | **+0.2** pts (31.7 vs 31.5) |

Extended from three New York rows to eight on 2026-08-17. The conclusion below was reached on three and is unchanged; it is now carried by every general in the file rather than by the two odd years that happened to be tabulated.

- **New York — the partisan asymmetry is real at the presidential level and concentrated in 2025 off-cycle.** At matched presidential salience New York's Republican electorate is 3.7 points older on the 65+ share where Idaho's is 0.2 — a genuine cross-state difference. But the eye-catching version, a 10.7-point gap with the Republican median voter at 62 against the Democratic 54, is **New York's 2025 general only**. Its 2023 general — the contest Finding 1 identifies as New York's genuinely lowest-salience — is nearly symmetric at 2.0 points. On the full eight-general set that contrast is now emphatic rather than suggestive. **Excluding 2025, New York's four other odd-year gaps run +1.0 to +3.7, against +2.7 to +4.1 in its three presidential years**: the ranges overlap and the presidential one sits slightly higher, so low salience does not widen the partisan age gap at all. 2025 is more than double the widest gap in any other general in the file. So "the Republican wing ages hardest as salience falls" does not hold across New York's own odd-years; what 2025 shows is that a *particular contest* mobilised Democrats young, not that low salience does this generally.

*What DOES hold, 8 times out of 8, is the level ordering* — the Republican electorate is the older of the two in every general New York's file records, and the no-party bloc is younger than both in every one. That is the durable claim, and it is the one the New York paper now makes. - **Idaho — senior representation is near-identical across the parties.** The Republican and Democratic electorates carry the same 65+ share (**31.7% vs 31.5%** in the 2024 general; medians 54 vs 50). The unaffiliated bloc sits ten points below both on that measure (21.3%), which is what pulls its median to 46. *Read that as a senior-end result only: on the under-30 share, unaffiliated (19.6%) and Democratic (19.4%) voters are level, and this bullet said Idaho's "youth lives outside the two major parties" until 2026-08-15, when an external referee showed that is false against the Idaho paper's own table.* Note that Idaho's own lowest-salience contest is the closed Republican primary, which is single-party by construction, so **no partisan-gap measurement is possible there at all**; the presidential row is the only matched comparison available. - **The closed primary is a friction, not an exclusion.** Idaho's *Democratic* primary is open to unaffiliated voters, and an unaffiliated elector may enter the Republican primary by affiliating at the poll book. In May 2024, 8,453 currently-unaffiliated voters pulled a Democratic ballot and 1,554 pulled a Republican one. An earlier version said the closed primary "locks that youth out by design"; the accurate statement is the Idaho paper's — an unaffiliated voter **must first affiliate** to vote in the contest that chooses the winner. - **Washington — age only, by construction.** No party of record; the WA paper deliberately makes no partisan claim, which is why it is the clean high-access baseline for the age result rather than a partisan one.

The lesson a single state cannot deliver: "low-salience electorates are old *and* skew Republican" is not a general fact. It fails in Idaho, where the graying is symmetric — and it fails within New York too, whose lowest-salience general is nearly as symmetric as Idaho's presidential one. The partisan asymmetry that does survive is smaller and sits at *high* salience (+3.7 against +0.2), plus one exceptional mayoral contest. The partisan structure therefore differs across these institutional settings (open vs closed primary; whether a party of record exists at all) — but this design does not isolate those rules as the cause. Three purposively chosen states, each differing on several dimensions at once, cannot separate a registration rule from everything else that distinguishes New York from Idaho. What the comparison establishes is that the age skew's partisan consequence is *not* a constant, which is the claim a single state cannot deliver.

## Finding 3 — Comparable representativeness yardsticks

The WA paper's two reusable cross-state metrics — the **age-dissimilarity index vs CVAP** (½·Σ|electorate-share − CVAP-share| over age bins) and the **habitual-core overlap** (share of a low-salience electorate that also cast a 2024 presidential ballot) — are designed to travel. They are now computed *identically* across WA/NY/ID by `scripts/diag_cross_state_age_harmonized.py`: shared age bins, age measured at each election date, and one ACS vintage for every state's benchmark (2020–2024 5-year, table B29001, pinned to `docs/reference/cvap_age_acs2024.csv` by `scripts/acs_cvap_by_state.py`).

The benchmark those indices are measured against, printed here because it is the instrument's denominator and appeared nowhere in earlier versions (ACS 2020–2024 5-year, table B29001, citizen voting-age population):

| CVAP share | 18–29 | 30–44 | 45–64 | 65+ |
|---|---|---|---|---|
| Washington | 19.8% | 26.7% | 30.9% | 22.6% |
| New York | 20.1% | 24.2% | 32.0% | 23.8% |
| Idaho | 21.3% | 25.5% | 30.2% | 23.0% |

| Metric | WA | NY | ID |
|---|---|---|---|
| Age-dissimilarity vs CVAP (pres → lowest-salience) | 7.5 → 18.6–20.0 | 7.0 → 11.8–22.4 | 8.2 → 25.0–27.8 |
| Habitual-core overlap (low-salience ⊂ 2024 presidential) | 95.5–97.5% | 86.4–95.6% | 94.4–97.8% |

**Every core cell measured BEFORE November 2024 is an upper bound, and the row is left uncorrected on purpose.** Each electorate is reconstructed from a roll built after the fact, so a voter who cast that ballot and has since died or moved is dropped from both sides of the overlap — and that is exactly the population which could not have voted in 2024. The bias grows with distance from the reference and is near zero for a contest at or after it. Washington is the only one of the three states with a retained roll snapshot, so it is the only place the size can be measured rather than argued: adding back the drop-offs it can still see moves 2021 from 95.5% to **89.9%**, 2023 from 97.5% to **95.0%**, and 2025 — four months from the roll — by **0.1** points. Those are themselves lower bounds. New York and Idaho have no equivalent snapshot, so correcting Washington's cell alone would put one column of a comparison row on a basis the other two are not on, which is the error this table exists to avoid; the cells stay comparable and the caveat is stated instead. **Read this row as a bound, not as a ranking.** An earlier version of this note said to read its ordering rather than its levels; that conflicted with Boundary of inference, which says

the cross-state ordering of low-salience cells is itself not robust to the lag confound, and the note was wrong to carve out an exception. What can be said is directional and does not need an ordering: New York's low endpoint is its **2017** contest, seven years before the reference and so the most inflated number here, while Idaho's is a **2026** primary that postdates the reference and is essentially unbiased — so correcting every cell would move New York's floor furthest and Idaho's barely at all.

**The dissimilarity index puts the three states in the same order the ratio did.** All three presidential electorates sit close to their citizen voting-age population (7.0–8.2 index points, i.e. ~7–8% of the distribution would have to move cohorts to match it). At lowest salience Washington reaches 18.6–20.0 and New York 11.8–22.4, while **Idaho's closed primaries reach 25.0–27.8 — about a quarter again (27.8 / 22.4 = 1.24) as far from the eligible population as the worst general election measured here.**

**That comparison is between each state's most extreme cell, and those cells sit at lags of nought to nine years — the confound Finding 1's own footnote says makes cross-state comparison of *levels* unsafe.** The claim is therefore restated on **lag-matched** cells: each state's low-salience contest nearest its roll extract, which puts all three on nearly one reconstruction basis.

| lag-matched cell | n | dissimilarity | 65+ |
|---|---|---|---|
| **ID** — May 2026 closed Republican primary | 247,391 | **27.8** | 48.6% |
| **WA** — Nov 2025 odd-year general | 1,993,489 | 18.6 | 40.3% |
| **NY** — Nov 2025 odd-year general | 4,039,099 | 11.8 | 34.4% |

Electorate sizes are printed here and in Finding 2 because no earlier version of this paper stated a single one, and a cross-state comparison of composition with invisible denominators asks the reader to take the precision on trust. Idaho's primary electorate is an order of magnitude smaller than the other two, which is itself part of the finding.

**On cells matched for lag, Idaho's closed primary is still the least representative low-salience electorate measured here, by 9.2 index points over Washington.** That is the claim this paper makes. It is narrower than a comparison of extremes, and unlike that comparison it is not retracted by the paper's own boundary section. (An earlier version said "a third again", which the two figures in the same sentence do not support. The ordering is the same on both metrics; the *gap between states* is narrower on the dissimilarity index than on the ratio, not wider, so an earlier "widens the gap" is also withdrawn.) The pattern is not that low-salience electorates are unrepresentative; it is that a *closed nominating* electorate is unrepresentative in a way that no general election in either other state approaches.

**The habitual-core figures say the same thing from the other direction.** Between 86% and 98% of every low-salience electorate also voted in the 2024 presidential general, so these are not different populations — they are the presidential electorate's standing core, with the peripheral voters stripped away. Idaho's 2024 primary is the *most* core-like (97.8%); New York's 2017 general is the least (86.4%).

**Read this metric with the lag in mind, because it is not the same construct across the set.** Every cell measures overlap with the *2024* presidential general, so a contest close to 2024 is being compared with itself over months while one far from it is compared over years, and attrition scales with that distance. Idaho's series makes the dependence visible: 94.7% (2022) → **97.8% (2024)** → 94.4% (2026), with the maximum at the primary nearest its own reference election, six months away. New York's runs 86.4 (2017) → 88.3 → 90.5 → 95.6 (2023) in order of recency. An earlier version read Idaho's 97.8% as showing "a selection result rather than a turnout-noise result"; the figure cannot carry that on its own, because the same-cycle proximity would raise it regardless. What survives is the weaker and still useful statement: every low-salience electorate in every state is drawn overwhelmingly from the presidential one, so none of them is a different population.

**Why the WA column is not character-identical to the WA paper's own ladder** (7.4 → 18.5–19.9). It differs by 0.05 index points and the cause is entirely the benchmark's precision: the single-state paper differences against its *published* CVAP row, rounded to one decimal, while the harmonized version uses the unrounded ACS shares. Computed against the paper's rounded row this script reproduces 7.4 / 13.2 / 18.5–19.9 exactly, and it asserts that on every run — so the WA column is a reconciliation of the two conventions, not a revision of the single-state result.

**And the same is true of the WA habitual-core column, by more than a rounding step** (recorded 2026-08-11). Its floor sits above the Washington paper's, which reads **92.2**% for the same three off-years, and both are asserted by their own verifiers. The gap is a population difference, not a disagreement: the WA paper counts every voter carrying a history record for the contest, while every cut in this study is age-banded and therefore joins the roll and requires a usable birth year. In 2021 that drops **105,969** voters whose overlap with the 2024 presidential is much lower, which lifts that year's figure by about three points; the other two off-years barely move. The harmonized column is the right one for cross-state comparison — it is the same population in all three states — and the single-state figure is the right one for Washington's own paper. As

with the dissimilarity ladder, neither is a revision of the other.

## Boundary of inference

- **Current-roll survivorship.** Past electorates are reconstructed from a *current* registration file, so voters who since died, moved, or were purged go missing. The WA paper benchmarks against certified counts and formally bounds the effect (the age result survives the worst case); NY and ID inherit the same caveat.

**This protects the direction, and only the direction.** Finding 1's second-order claims — the magnitude spread, and the cross-state *ordering* of low-salience levels — are not robust to it, because the reconstruction bias runs with lag and the states' cells sit at different lags (see the ladders under Finding 1: NY 28.4 → 41.6 across 2017–2023, ID 41.4 → 48.6 across 2022–2026, both near-monotone in recency). Worse, the direction of the bias need not match across states: Washington's attrition is older-skewing, so correcting it would raise its senior share (the direction its own paper measures and bounds), while Idaho's churn is largest among the young, unaffiliated and movers — a profile *consistent with* over-retaining seniors, though the Idaho paper treats the net direction of its bias as unestablished and relies on arithmetic missing-voter bounds instead, and this paper follows it. If the consistent-with reading is right, the state supplying the study's extreme is the one whose reconstruction most over-retains seniors relative to youth. Nothing here tests that.

**This is why Finding 3 no longer rests on a comparison of extremes.** An earlier version stated the headline on each state's most extreme cell and then retracted the basis for it here, which left the paper arguing both sides in different sections. Finding 3 is now made on **lag-matched** cells — each state's low-salience contest nearest its roll extract — and the ordering survives that restriction with Idaho ahead by 9.2 index points. The comparison of extremes stays in the text as context and is explicitly not the claim. - **Idaho rates are unusable, and shares are not immune.** The roll contraction inflates ID turnout rates, so Idaho is composition-shares only and all cross-state comparisons are share-based. That does not neutralise the problem: the Idaho paper states the bias is *larger* for high-churn groups — the young, the unaffiliated, movers — and a bias larger for the young **is** a bias in the age composition. Switching to shares removes the rate artifact, not the differential one. - **Party of record is a 2026 snapshot in both party states, and Idaho's is reactive.** New York's enrollment and Idaho's party are read from a current extract, not as of each analysed election, so every Finding 2 figure describes *today's* partisans' past behaviour. Both companions flag this as their largest limitation. Idaho's case is worse than a staleness problem: under Idaho Code § 34-411A an unaffiliated elector who requests a Republican primary ballot is thereafter registered Republican — subsection (2) directs the county clerk to record the poll-book affiliation into the elector's registration record after the primary — so the snapshot mechanically reclassifies the very voters whose absence Finding 2 discusses, and the Idaho paper reports its unaffiliated primary shares as **lower bounds** for that reason. - **Three states, and what the third axis would have added.** The set varies party of record (WA no; NY and ID yes) and openness of the lowest-salience measured contest (ID's closed primary against three open generals). It does **not** contain a state that lacks a party of record yet publishes, durably, which party's primary each voter pulled — the configuration that would let Finding 2's partisan question be asked where party is unrecorded. **Texas is that case** (no party registration; primary participation is public record), and it is future work rather than a gap here. Washington is a near-miss worth naming rather than a counterexample: its presidential primary *does* produce a per-voter party declaration, but it is publicly disclosable only inside the post-primary window of RCW 29A.84.730, the 2024 window has closed, and only Pierce County was captured in time — so the asset exists in principle and not in hand. Loading Texas would test whether Washington's silence on party is a data limit or a real absence of partisan structure, and it would add the one large, low-access, Southern case the set otherwise lacks. Until then, Finding 2 is stated as a claim about **party-registration states**, which is what the data supports, and Finding 1's recurrence claim rests on three states rather than four. - **Comparability is bounded by the protocol, not eliminated.** The lowest-salience class is genuinely different across states; the paper's claim is about the *gradient within each state's own salience ladder*, harmonized on the two directly-comparable classes.

## What it means

Across three states spanning the partisan and institutional map, the same structural fact holds: the lower the salience, the older and smaller the electorate that turns out.

The inference from that to *timing as the lever* is weaker than it reads, and is stated here at the strength it can carry. The presidential electorates being nearly identical is close to mechanical — three states with similar CVAP age structures and near-ceiling senior turnout at high salience will converge at the top of any such ladder — so the convergence confirms a weak prior rather than identifying a mechanism. More importantly the three low-salience classes are **not the same treatment**: Washington's is a local general, New York's is a local general that in three of five years contains a New York City mayoral race, and Idaho's is a closed party primary. Contest content varies with the state, so divergence across them cannot

isolate timing. What the data support is the weaker and still useful statement that **every one of these low-salience contests is older than its own state's presidential electorate**, which is a within-state comparison and immune to the objection.

The partisan consequences are not portable either: they vary with institutional setting, and this design cannot say which feature of that setting produces the variation. On the policy implication the WA paper draws — moving low-salience local contests on-cycle — this study shows the *composition* problem it addresses is present in all three states; it does not show the remedy transfers, and Ornstein (2024) found the turnout and diversity gains without a detectable downstream effect on representation or policy. The assumption that lower-salience electorates tilt uniformly toward one party does not survive at all.

## Methods & reproducibility

**Findings 1 and 3 come from one script, not from three papers.** `scripts/diag_cross_state_age_harmonized.py` — the age twin of `scripts/diag_cross_state_giving_turnout.py` — reads `data/{wa,ny,id}_vrdb.duckdb` read-only and computes every cell in both tables with the same age bins, the same band boundaries, and age measured at each election's own date. It emits aggregates only and never a row; the three files are used under RCW 29A.08.720, New York FOIL lawful-use terms, and Idaho Code § 34-437A respectively.

Three conventions in that script are load-bearing, and each was measured rather than assumed:

- **All three states are year-of-birth resolution, but not on the same clock — and the difference is worth two points of the 65+ share.** WA and NY publish a birth *year*, normalised to July 1 — every row, verified — so `date_diff('year', …)` against a November election is the calendar-year difference, which implicitly assumes the birthday has already happened. Idaho publishes a current integer *age*, which already accounts for whether it has, and `age − (2026 − year)` inherits that. The two conventions therefore disagree by a year for every Idahoan whose 2026 birthday falls after the extract date, and an integer age cannot say who that is.

So Idaho's figure is a bracket, not a point, and **the bracket is one-sided**: `age` is its low end and `age + 1` its high. Measured across Idaho's five classes it is **1.7–2.6 points of the 65+ share** — more than it sounds, because one single-year cohort near 65 is about 1.8% of the Idaho roll — and **1.3–1.4 points of the dissimilarity index**. The figures reported throughout this paper are the low end, so Idaho's position is stated conservatively: **the May 2024 primary's** dissimilarity is 27.6 on the convention used here and 29.0 on the other end. *(That names one cell rather than the class, and the distinction matters because the tables above report the primaries' range as 25.0–27.8 — the 2024 cell is 27.6 and the 2026 cell 27.8, so an undeclared "the closed-primary electorate's dissimilarity" reads as the range's top and is a different number. Scope stated 2026-08-11.)* There is nothing to correct to, because the point estimate is not recoverable from an integer age — but the correction runs one way, and it runs toward the finding rather than against it. `scripts/diag_cross_state_age_harmonized.py` recomputes the bracket on every run and refuses to finish if it stops being one-sided. - **Idaho's lowest-salience class is selected on the ballot the voter actually pulled** ( `ballot_choice = 'REP'` ), not on party of record. That is the field that makes Idaho's exclusion a measurement rather than an inference, and no other state in the set has it. - **The CVAP benchmark is pinned, not fetched.** `scripts/acs_cvap_by_state.py` writes `docs/reference/cvap_age_acs2024.csv` (ACS 2020–2024 5-year, table B29001) for all three states at one vintage; the harmonizer refuses to run without it. A per-run fetch would let one state's benchmark move to a newer release while the others stayed, and the dissimilarity index would quietly stop being comparable.

`derive()` **reads the dissimilarity ladder out of** `who-decides-washington.md` and raises unless its own definition reproduces that ladder exactly at the paper's printed precision — on the paper's own rounded benchmark, so the test isolates the definition rather than the benchmark's precision. It runs inside `derive()` rather than in the script's `main()`, which is what makes "on every run" true: every automated consumer, including this paper's verifier, calls `derive()`. A missing anchor is a failure, not a skip, so rewording the sentence out from under the check trips it. Both failure modes — a changed ladder in the paper and a drifted definition here — are mutation-tested in `tests/test_infrastructure/test_harmonizer_wa_crosscheck.py`.

An earlier version of this paragraph described a guard that did not exist as described: the check sat in `main()`, which no automated path called; it compared against hardcoded literals and never opened the WA paper, so it could not see the decoupling it was said to prevent; and it signalled by return code. All three are fixed, and the mutation tests exist so that "this guard works" is a tested claim rather than a written one.

## Related work

- **Turnout composition by salience (surge-and-decline).** Campbell, "Surge and Decline" (1960); Wolfinger & Rosenstone, *Who Votes?* (1980); Leighley & Nagler, *Who Votes Now?* (2013); Plutzer, "Becoming a Habitual Voter"

(2002).

- **Off-cycle election timing, composition, and representation.** Anzia, *Timing and Turnout* (2014); Hajnal & Trounstine (2005); Hajnal, Kogan & Markarian (2022); Kogan, Lavertu & Peskowitz (2018); Einstein, Palmer, Hamilton & Singer, "Age and Homeownership Drive the Local Turnout Gap," *Urban Affairs Review* (2025); Lucero, Robles, Trounstine & Collins, "What Date Works Best?" (2025).
- **Contested policy effects of timing.** Ornstein, "Election Timing Revisited" (2024) — finds the expected turnout and diversity gains from California's on-cycle mandate but **no** detectable downstream effect on representation, incumbency or policy. Cited here because this paper's "What it means" section reaches a policy implication and the companion Washington paper treats this null as the main counterweight to it; dropping it from the list while keeping the implication would be selective.
- **The primary as the real election under one-party dominance.** V.O. Key, *Southern Politics in State and Nation* (1949); Hirano & Snyder, *Primary Elections in the United States* (2019).
- **Primary-electorate representativeness.** Sides, Tausanovitch, Vavreck & Warshaw, "On the Representativeness of Primary Electorates" (2020).
- **Independents / the unaffiliated.** Klar & Krupnikov, *Independent Politics* (2016).
- **Voter-file / individual-level method.** Ansolabehere & Hersh, "Validation…" (2012); Hersh, *Hacking the Electorate* (2015); Feder & Miller, "The Racial Burden of Voter List Maintenance Errors," *Science Advances* (2020).

## Status / next actions

- [x] Harmonization protocol (comparable election classes per state).
- [x] Finding 2 party contrast.
- [x] **Harmonized-metrics script written** (2026-08-06): `scripts/diag_cross_state_age_harmonized.py` + `scripts/acs_cvap_by_state.py`. Findings 1 and 3 now come from one code path; Finding 3's placeholders are filled. Filling them changed two things in Finding 1's prose, both recorded there: the senior-to-youth ratio does **not** "roughly triple in every state", and Idaho's lowest-salience cell became a measured value rather than the phrase "grayest of all". *(This bullet quoted that day's figures — a 1.5×–4.9× multiplier and a single Idaho cell of 46.7% / 5.0% — and both were superseded within the same round, when the low-salience column was widened to use every such contest on file (3 / 5 / 3 rather than 3 / 2 / 1). Finding 1 now carries the current values: the multiplier runs 1.5× to 5.1× and Idaho's cell is a range. Corrected here 2026-08-11, because a status bullet reads as current fact.)*
- [x] **TX decision made 2026-08-06: publish as a three-state paper.** Texas moves from a deferred fourth case to named future work in Boundary of inference. The reasoning is on the record in `docs/electoral-health-audit-log.md`: the paper's claim is about the gradient within each state's own salience ladder, three states already vary both institutional axes it turns on, and a Texas primary-history *proxy* is not the same measurement as party of record — it would need its own validation rather than slotting into the existing tables. It is a good next paper, not a blocker for this one.
- [x] **Submission metadata + notes written** (2026-08-06): `who-returns-ballot-submission-metadata.md` and `-notes.md`, registered in `scripts/check_cross_doc_consistency.py` the day they were written.
- [ ] Independent human verification pass (per `electoral-health-audit-log.md`) before external review; target venue State Politics & Policy Quarterly. **This is now the only blocker.**
- [ ] Future work, not blocking: load the Texas voter file and add a primary-history party proxy, with the validation that a proxy requires. See Boundary of inference.